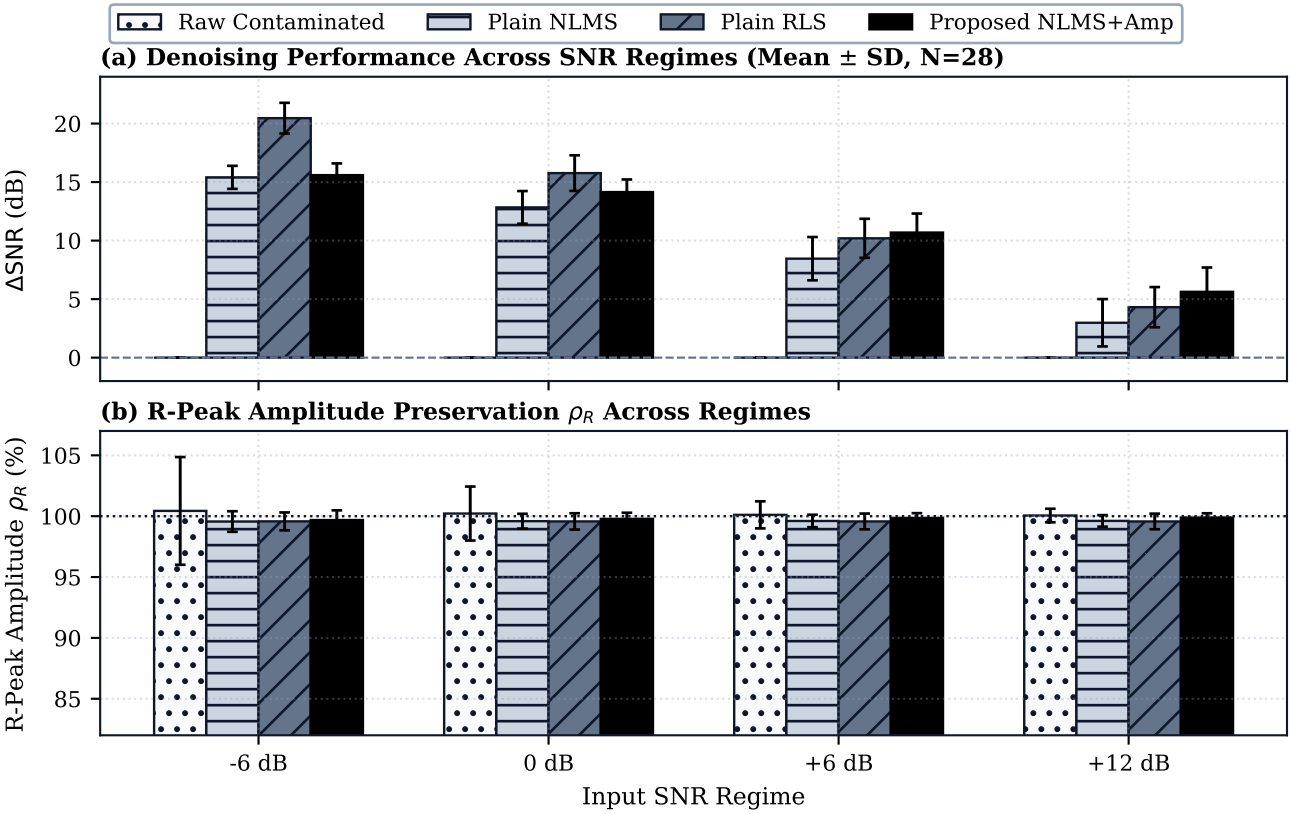
Raw Contaminated
Plain NLMS
Plain RLS
Proposed NLMS+Amp
(a) Denoising Performance Across SNR Regimes (Mean ± SD, N=28)
ΔSNR (dB)
20
15
10
5
0
(b) R-Peak Amplitude Preservation ρ_R Across Regimes
R-Peak Amplitude ρ_R (%)
105
100
95
90
85
-6 dB
0 dB
+6 dB
+12 dB
Input SNR Regime